

STAIR

Scaffold for Teaching Agent
Integration Readiness

A Five-Level Framework for Evaluating LMS Readiness for Agentic AI Teaching Agents

We don't need smarter AI. We need LMSs ready to host it.

The ceiling is set by host architecture, not model capability.

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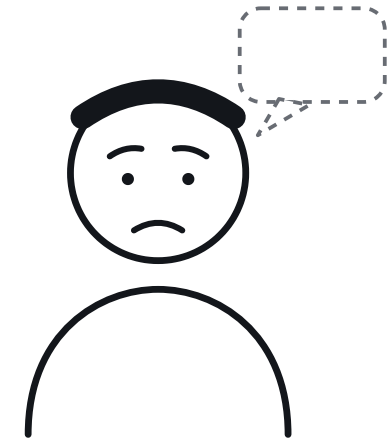
Picture a second-year. Call him Tom.

It is 02:47. A learner like Tom has just failed quiz 2 a second time, and closes the laptop. The course has an AI tutor switched on. It says nothing. Nothing reaches him.

Tom is an illustrative composite, not a study participant; every beat is something a real platform can or cannot do.

This is a **buying decision**, not a tech-support problem.

Why couldn't it help him?



TOM

02:47 · the tutor stays silent

CONTEXT

WHAT AN AI TEACHING AGENT IS

A colleague that acts. Not a chatbot you open.

A CHATBOT

You open it and ask.

Reactive. It waits for a student to come to it. Most "AI in the LMS" today is this.

A COLLEAGUE THAT ACTS

It notices, decides, delivers.

It sees a learner like Tom slipping, decides he needs a nudge now, and delivers it inside his course, on its own initiative.

Our LMSs were built to deliver content and record activity, **not to act like colleagues**. That gap is architectural, not a setting you switch on.



Tom · 02:47 · no agent watching | ILLUSTRATIVE

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PURPOSE & RESEARCH QUESTIONS

ONE RULER, ONE TEST

Two questions.

RQ1 / THE RULER

What does an LMS need, in its architecture, to host an AI tutor?

to be answered

RQ2 / THE TEST

Does Moodle have it?

to be answered

Moodle is just the worked example, because its source is open and I can read it. **The ruler scores any LMS.** If you run Canvas or Blackboard, this is about you too.



STAIR: five levels.

L1	Sense	See what a learner like Tom does, as it happens.	02:47
L2	Remember	Hold his whole trajectory: grades, progress, risk.	02:49
L3	Reason	Decide he needs help now, within pedagogical limits.	02:51
L4	Coordinate	Several helpers working together, not over each other.	02:53
L5	Reach	Get help to him, in context, before he closes the laptop.	02:55 laptop closes

Tom's clock runs down the right of the ladder: 02:47 to 02:55, when the laptop closes. Each level is a minute the platform has to climb. (illustrative)



Tom · each level is something he needs | ILLUSTRATIVE



TOM
02:47 · the climb
begins

Three literatures converge on the same five levels.

■ Each level is grounded by a matching construct in self-regulated learning, multi-agent systems, and educational LLM agents.

STAIR LEVEL	ZIMMERMAN: SELF-REGULATED LEARNING	TALEBIRAD & NADIRI: MULTI-AGENT	CHU ET AL.: EDUCATIONAL AGENTS
L1 Sense	Self-observation	Event sensing layer	Contextual input
L2 Remember	Self-judgment	Knowledge representation	Memory, short and long
L3 Reason	Strategic planning	Agent reasoning loop	Planning and goal decomposition
L4 Coordinate	Metacognitive monitoring	Multi-agent coordination	Multi-agent communication
L5 Reach	Adaptive self-reactions	Delivery mechanism	Personalisation and adaptation

Three fields, one structure. And each level needs the one below it: you can't reason about a learner you can't remember.

GROUNDING IN

1. Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory into Practice*, 41(2), 64–70.

2. Talebirad, Y., & Nadiri, A. (2023). Multi-agent collaboration: Harnessing the power of intelligent LLM agents. *arXiv:2306.03314*.

3. Chu, Z., et al. (2025). LLM agents for education: Advances and applications. *Findings of the ACL: EMNLP 2025*, 13782–13810.



METHODOLOGY

THE DATA FLOOR, 02:47 TO 02:49

L1-L2: perceive and remember.

A LEARNER LIKE TOM ILLUSTRATIVE

Activity happens

- opens the course, 02:47
- quiz 2, still not submitted
- third late-night login this week

L1 SENSES

L1 SENSE

Each becomes an event

The platform can see every action the moment it happens.

```
login · 02:47  
quiz_2 = not_submitted  
logins this week ↑
```

L2 REMEMBERS

L2 REMEMBER

Events combine into a state

Held together, the events become a picture it can keep.

engagement falling · risk rising

a model of a learner like Tom, not a fact about him

These two levels are about **data**: the floor every later level stands on. The platform can build this picture. Watch what it can do with it.



TOM

the platform can sense
him



Tom · 02:49 · risk could be modelled | ILLUSTRATIVE

METHODOLOGY

THE AGENT LAYER, 02:55

L3-L5: reason, coordinate, reach.

L3 REASON

Decide what would help

Take the state from L2 and conclude a nudge is needed now.

HOLD THIS LEVEL IN MIND



L4 COORDINATE

Line up the helpers

Several agents act together, not over each other.



L5 REACH

Deliver in the moment

Get the nudge to a learner like Tom, in context, before the laptop closes.

02:47 risk emerges

02:55 · laptop closes, nothing sent

In this scenario, that is the gap the platform cannot cross. (illustrative scenario)



TOM

02:55 · still nothing sent



Tom · 02:55 · laptop closing, nothing sent | ILLUSTRATIVE

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Two artefacts, one open platform.

01 PROBLEM

No instrument

Nothing measures whether an LMS can host an autonomous teaching agent.

02 BUILD

Two artefacts

The STAIR framework (a model) and the scoring instrument (a method).

03 APPLY

Score Moodle

Run the instrument over Moodle 5.2dev+ by source-code analysis.

04 EVALUATE

Against evidence

Every score tied to the code, reproducible from the public repo.

Design Science Research (Hevner; Peffers). Why Moodle:

Most deployed.

The open-source LMS used most worldwide.

First with native AI.

A dedicated `core_ai` subsystem.

Fully readable.

Source open, so every claim is verifiable.

28 checks, scored against evidence.

EACH CHECK: A 0 TO 4 SCORE

0	Absent	0%
1	Nascent	25%
2	Partial	50%
3	Substantial	75%
4	Ready	100%

Evidence hierarchy:
Source code (primary) > API / web-service > runtime > docs.
A zero requires a source search proving absence.

FROM 28 CHECKS TO A LEVEL SCORE

28 sub-dimensions across the five levels (5 / 6 / 7 / 5 / 5).
They roll up, **equally weighted**, into a readiness percentage for each level. No basis yet for weighting one part above another.
Every score is pinned to a line in Moodle's source, so **anyone can check it**, including someone who has never read code. Full provenance in `scoring_provenance.json`.



FINDINGS

SO FAR, SO GOOD

We scored Moodle.

55

L1 SENSE

53

L2 REMEMBER

?

L3 REASON

Just over half on both. It can partially sense and model someone in Tom's position.

Now, level 3: decide a learner needs help, and act. How ready, 0 to 100?

Pick a number. Hold onto it.



Tom · seen and modelled, not helped | ILLUSTRATIVE

FINDINGS

THE 28-POINT DROP

Readiness doesn't decay. It collapses.

A **28-point** cliff opens between data (L2, 53) and reasoning (L3, 25). The fall is a discontinuity, not a slope.

Why: the platform can answer with a language model, and separately flag who is at risk. The two were never wired to talk to each other.

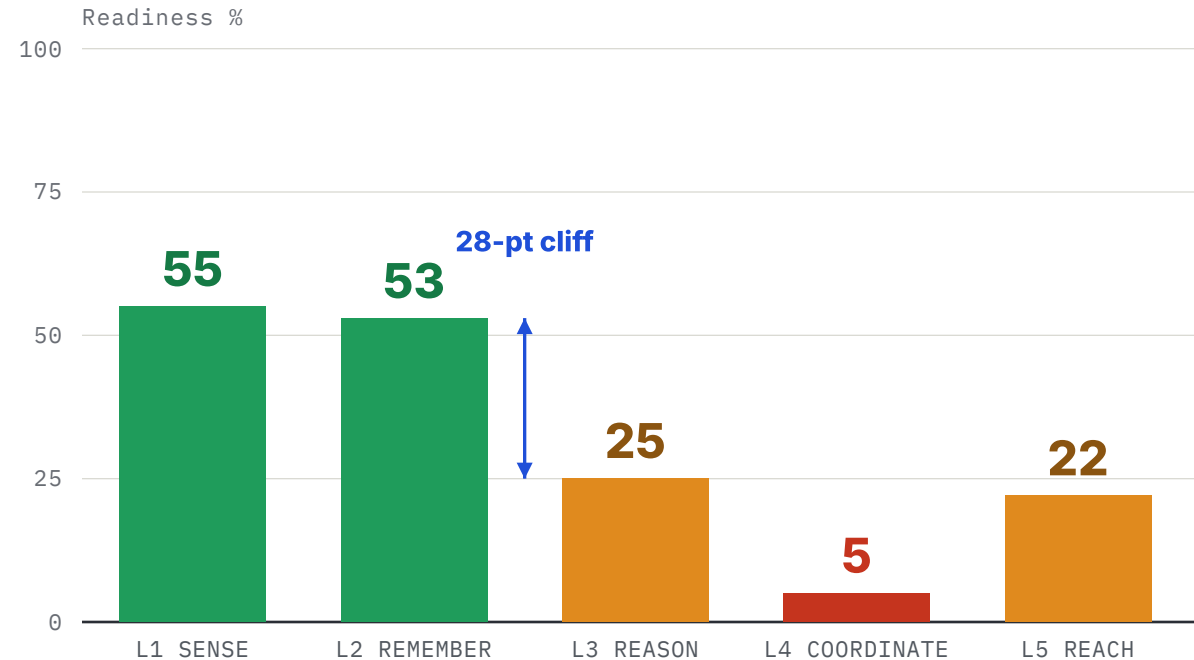
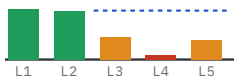


Fig. STAIR readiness, Moodle 5.2dev+ · source-code structural analysis

Claim: no automatic pathway from prediction to action.
Not a claim that Moodle cannot help learners by other means.



Tom · the cliff is where help stops | ILLUSTRATIVE

FINDINGS

THE MECHANISM, WHY IT COLLAPSES

Two rooms. No door between them.

THE HELP, READY TO SEND

ROOM A · AI/

Can draft a tailored message

The AI subsystem can generate a genuinely good, personalised response.

L3 Reason: no link

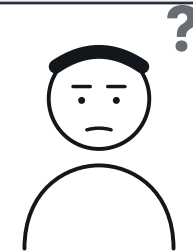
THE STATE, FROM L1-L2

ROOM B · ANALYTICS/

Can flag who is at risk

The analytics framework can identify a learner like Tom as at risk.

The platform can speak. It can see who is drowning. It can't connect the two.



TOM

seen, but not reached

FINDINGS

EVERY SCORE TRACED TO CODE

Why 25, in the numbers.

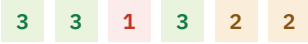
L1 Sense

5 checks



L2 Remember

6 checks



L3 Reason

7 checks



L4 Coordinate

5 checks



L5 Reach

5 checks



each cell is one of the 28 checks, scored 0 (absent) to 4 (ready). The zeros cluster at the agent layer, L3 and L4. Per-level readiness is the equal-weighted mean of five capability dimensions (shown for L3 at right), not a raw average of these cells.

L3 = the mean of five

its 7 checks roll into these 5 capability dimensions

Coverage	50
Accessibility	50
Educational specificity the reasoning layer	0
Temporal	25
Integration knowing-to-acting wire	0

L3 readiness 25

These two zeros open the 28-point cliff: no agent reasoning layer, no AI-analytics bridge. No honest weighting scores around them.



DISCUSSION

GENERALISABILITY, RQ2 ANSWERED

This stops being a Moodle problem.

■ RQ2 answered: Moodle clears the data floor, then collapses at reasoning.

I could only measure Moodle because its code is open.

Canvas and Blackboard share the era-defining pattern: a content-delivery core with analytics bolted on as a separate reporting layer. That is the design that produces the wall, so it is very likely there too, you just cannot open them to check. STAIR is how you would confirm it. There is a learner like Tom in every cohort, on every one of these platforms.

Moodle measured: the cliff is real	Canvas same era, likely the same wall	Blackboard sealed, you cannot check inside
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Below the cliff: L4 Coordinate 5, L5 Reach 22. The delivery pipes exist; nothing upstream decides when to use them.



Tom · a learner like him in every cohort | ILLUSTRATIVE

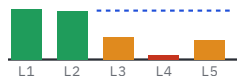
IMPLICATIONS

THE FIX, THE SURPRISING PART

One doorway.



One small plugin wires the risk prediction into the AI's action context, `get_predictions` → AI action: an estimated **150-200 lines** from source analysis, not yet built, no change to Moodle's core. **Nobody has been asked to build the door yet:** it sits across two subsystems, `ai/` and `analytics/`, that no single team owns.



Tom · help: possible | ILLUSTRATIVE



TOM

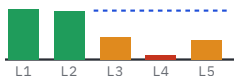
a message can
reach him

Three questions for your LMS.

- 1 Can it sense what learners do, as they do it?
- 2 Can an agent read a learner's full state?
- 3 Can support reach the learner, in context, at the right moment?

Audit your stack before you buy more AI to bolt onto it.

For procurement: require suppliers to **demonstrate an automatic pathway from at-risk prediction to learner-facing action**, and to publish their STAIR profile. The wire is small; the policy around an autonomous agent, the DPIA and accountability, is the real institutional work, and STAIR shows you where to scope it.



DISCUSSION

WHAT THIS DOES AND DOES NOT CLAIM

The bounds, stated plainly.

ONE RATER

Every score is **anchored to source code**, so it is reproducible. The level mapping and the scores are one author's coding, pending a **second rater**.

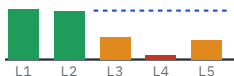
STAIR measures structural **capacity to host** an agent, not learning outcomes. A high score enables good teaching; it does not guarantee it.

STRUCTURAL, NOT RUNTIME

These numbers are a **ceiling**. Runtime behaviour can only make them lower, never higher.

ONE PLATFORM, ONE MOMENT

Canvas and Open edX are next. That comparison is what will **calibrate the instrument**.



Structural readiness / a ceiling, not a floor

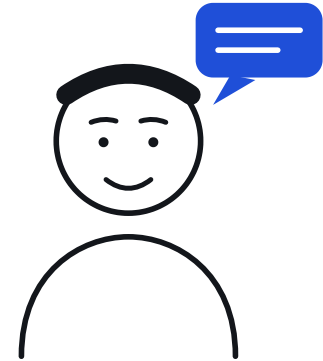
KEY TAKEAWAYS

BACK TO 02:47

Why it couldn't help him.

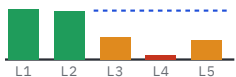
Not because the AI was too dumb. Because nothing connected what the platform could see to what it could say.

And that, we can fix.



TOM

help: now possible



Tom · help: none → possible | ILLUSTRATIVE

We don't need **smarter AI**. We need LMSs ready to host it.

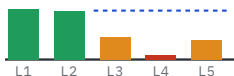
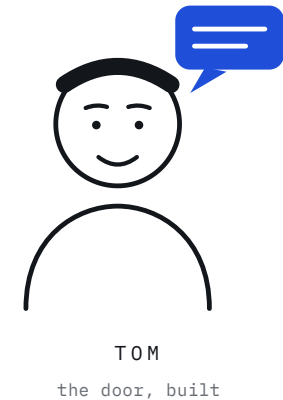
I'm building the **infrastructure and operating system for AI teaching agents**.

If you work in this space, I would love to collaborate on the research. Come find me.

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github.com/baqarjafri/paper-stair-framework · full rubric and provenance in the repo



Thank you / questions welcome

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